

ASSIGNMENT 7

Question: - 1

Create a class NumberSeries that prints numbers from 1 to 200 using a loop.

Ans: - Code

```
Question 1 > ...  
1 class NumberSeries:  
2     def print_numbers(self):  
3         for i in range(1, 201):  
4             print(i)  
5 num = NumberSeries()  
6 num.print_numbers()
```

Output: -

```
PS D:\Data Science with AI & ML\PYTHON\Assignment 7>  
hon311\python.exe' 'c:\Users\LENOVO\.vscode\extension  
ebugpy\launcher' '50258' '--' 'D:\Data Science with A  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11
```

```
191  
192  
193  
194  
195  
196  
197  
198  
199  
200  
PS D:\Data Science with AI & ML\PYTHON\Assignment 7>
```

Question: - 2

Create a class PrimeChecker with a function to check whether a number is prime or not.

Ans: - Code

```
class PrimeChecker:
    def check_prime(self, x):
        if x > 1:
            for i in range(2, x):
                if x % i == 0:
                    print(x, "is not a Prime Number")
                    break
            else:
                print(x, "is a Prime Number")
        else:
            print(x, "is not a Prime Number")
obj = PrimeChecker()
n = int(input("Enter a number: "))
obj.check_prime(n)
```

Output: -

```
Enter a number: 5
5 is a Prime Number
PS D:\Data Science with AI & ML\PYTHON\Assignment 7>
```

```
Enter a number: 6
6 is not a Prime Number
PS D:\Data Science with AI & ML\PYTHON\Assignment 7>
```

Question: - 3

Create a class Student using constructor to store:

name

roll number

marks

Ans: - Code

Output: -

```
Enter Student Name: Ashwani Kumar  
Enter Roll Number: 2  
Enter Marks: 90  
>>>>>>>>>>>>><<<<<<<<<<<<<<  
Student Name: Ashwani Kumar  
Roll Number: 2  
Marks: 90  
PS D:\Data Science with AI & ML\PYTHON\Assignment 7>
```

Question: - 5

Create a class Calculator with functions:

add()

subtract()

multiply()

divide()

Ans: - Code

```
class Calculator:
    def add(self, a, b):
        print("Addition =", a + b)
    def subtract(self, a, b):
        print("Subtraction =", a - b)
    def multiply(self, a, b):
        print("Multiplication =", a * b)
    def divide(self, a, b):
        if b != 0:
            print("Division =", a / b)
        else:
            print("Division by zero is not possible")

cal = Calculator()
n1 = int(input("Enter first number: "))
n2 = int(input("Enter second number: "))
cal.add(n1, n2)
cal.subtract(n1, n2)
cal.multiply(n1, n2)
cal.divide(n1, n2)
```

Output: -

```
Enter first number: 20
Enter second number: 10
Addition = 30
Subtraction = 10
Multiplication = 200
Division = 2.0
PS D:\Data Science with AI & ML\PYTHON\Assignment 7>
```

Question: - 6

Create a parent class Person and child class Employee using single inheritance. Display employee details.

Ans: - Code

```
class Person:
    def get_person(self):
        self.name = input("Enter Name: ")
        self.age = input("Enter Age: ")
class Employee(Person):
    def get_employee(self):
        self.emp_id = input("Enter Employee ID: ")
        self.salary = input("Enter Salary: ")
    def display(self):
        print("\nEmployee Details")
        print("Name:", self.name)
        print("Age:", self.age)
        print("Employee ID:", self.emp_id)
        print("Salary:", self.salary)
emp = Employee()
emp.get_person()
emp.get_employee()
emp.display()
```

Output: -

```
Enter Name: Ashwani Kumar
Enter Age: 22
Enter Employee ID: 102
Enter Salary: 20000

Employee Details
Name: Ashwani Kumar
Age: 22
Employee ID: 102
Salary: 20000
PS D:\Data Science with AI & ML\PYTHON\Assignment 7>
```

Question: - 7

Create a class BankAccount with functions:

deposit()

withdraw()

check_balance()

Ans: - Code

```
class BankAccount:
    def __init__(self):
        self.balance = 0
    def deposit(self):
        amount = int(input("Enter deposit amount: "))
        self.balance += amount
        print("Money Deposited")
    def withdraw(self):
        amount = int(input("Enter withdraw amount: "))
        if amount <= self.balance:
            self.balance -= amount
            print("Money Withdrawn")
        else:
            print("Insufficient Balance")
    def check_balance(self):
        print("Current Balance:", self.balance)
acc = BankAccount()
acc.deposit()
acc.withdraw()
acc.check_balance()
```

Output: -

```
Enter deposit amount: 32000
Money Deposited
Enter withdraw amount: 2000
Money Withdrawn
Current Balance: 30000
PS D:\Data Science with AI & ML\PYTHON\Assignment 7>
```

```
Enter deposit amount: 25000
Money Deposited
Enter withdraw amount: 25001
Insufficient Balance
Current Balance: 25000
PS D:\Data Science with AI & ML\PYTHON\Assignment 7>
```

Question: - 8

Create a class FactorialCalculator with a function to calculate factorial of a number using a loop.

Ans: - Code

```
class Factorial:
    def factorial(self, n):
        fact = 1
        for i in range(1, n + 1):
            fact = fact * i
        print("Factorial =", fact)
obj = Factorial()
x = int(input("Enter a number: "))
obj.factorial(x)
```

Output: -

```
Enter a number: 6
Factorial = 720
PS D:\Data Science with AI & ML\PYTHON\Assignment 7>
```